

Deep Learning

Complete Important Topics + Intuition + Interview WHY Questions

Goal: Focus on concepts that matter for ML/DL interviews and real projects. For each topic, understand what it is, why it is needed, and what an interviewer may ask.

1. Neural Network Fundamentals ■■■■■■

Artificial Neuron, Weights and Bias

Intuition: A neuron computes a weighted combination of inputs plus a bias and then applies an activation.

Interview WHY: Why are weights and bias needed? What does a neuron learn?

Layers and Network Architecture

Intuition: Layers transform representations step by step from raw input to useful features.

Interview WHY: Why do we need hidden layers? Why can deeper networks learn more complex functions?

Forward Propagation

Intuition: Input passes through the network to produce a prediction.

Interview WHY: What happens during a forward pass?

Loss Function

Intuition: Measures how different the prediction is from the target.

Interview WHY: Why can't we train a model without a loss function?

Backpropagation

Intuition: Calculates how each parameter contributed to the loss using gradients and the chain rule.

Interview WHY: Why do we need backpropagation? How does the error reach earlier layers?

2. Activation Functions ■■■■■■

ReLU

Intuition: $\max(0, x)$. It adds non-linearity and generally provides better gradient flow than sigmoid/tanh in deep hidden layers.

Interview WHY: Why is ReLU commonly used? What is the dying ReLU problem?

Sigmoid

Intuition: Maps values to 0–1 and is useful for binary probability outputs.

Interview WHY: Why is sigmoid often used in binary classification output? Why can it cause vanishing gradients?

Tanh

Intuition: Maps values to -1 to 1 and is zero-centered.

Interview WHY: Why might tanh be preferable to sigmoid in some hidden-layer settings?

Softmax

Intuition: Converts logits into a probability distribution across classes.

Interview WHY: Why use softmax for multiclass classification?

GELU / Leaky ReLU

Intuition: Know their basic intuition and why alternatives to ReLU exist.

Interview WHY: Why use GELU in Transformer-style networks? Why Leaky ReLU?

Non-linearity

Intuition: Without nonlinear activations, stacked linear layers collapse into one linear transformation.

Interview WHY: What happens if a neural network has only linear activation functions?

3. Loss Functions ■■■■■■

MSE / MAE / Huber

Intuition: Regression losses quantify prediction error; each reacts differently to outliers.

Interview WHY: Why might MAE be preferred over MSE? When is Huber useful?

Binary Cross Entropy

Intuition: Measures error between binary labels and predicted probabilities.

Interview WHY: Why use cross-entropy instead of MSE for classification?

Categorical Cross Entropy

Intuition: Used for multiclass classification with class probabilities.

Interview WHY: Why does cross-entropy work well with softmax?

4. Gradient Descent ■■■■■■

Gradient

Intuition: The gradient indicates the direction of steepest increase of the loss.

Interview WHY: Why do we move opposite to the gradient?

Learning Rate

Intuition: Controls the size of each parameter update.

Interview WHY: What happens if the learning rate is too high or too low?

Batch, SGD and Mini-batch Gradient Descent

Intuition: Different methods use different amounts of data for each parameter update.

Interview WHY: Why is mini-batch training commonly used?

Local vs Global Minimum

Intuition: Understand optimization intuition rather than only definitions.

Interview WHY: Can gradient descent get stuck? Why is non-convex deep learning optimization different?

5. Optimizers ■■■■■■

SGD

Intuition: Basic gradient-based parameter update.

Interview WHY: Why might SGD be slower or noisier than adaptive optimizers?

Momentum

Intuition: Uses previous update direction to smooth and accelerate movement.

Interview WHY: How does momentum help optimization?

RMSProp

Intuition: Adapts update size based on recent squared gradients.

Interview WHY: Why can adaptive learning rates help?

Adam

Intuition: Combines momentum-like first-moment tracking with adaptive second-moment scaling.

Interview WHY: Why is Adam widely used? What are its limitations?

AdamW

Intuition: Separates weight decay from Adam's gradient update.

Interview WHY: Why is AdamW preferred over naive L2 implementation with Adam?

6. Backpropagation and Chain Rule ■■■■■

Chain Rule

Intuition: Allows gradients to be propagated through a composition of functions.

Interview WHY: Why is the chain rule central to neural-network training?

Gradient Flow

Intuition: Gradients move from the loss backward through layers.

Interview WHY: What determines whether early layers receive useful gradients?

7. Batch Size, Epochs and Iterations ■■■■

Batch Size

Intuition: Number of training examples used for one parameter update.

Interview WHY: How does batch size affect memory, speed, and gradient noise?

Epoch

Intuition: One complete pass through the training dataset.

Interview WHY: What is the difference between an epoch and an iteration?

8. Overfitting and Regularization ■■■■■

Overfitting

Intuition: Model learns training-specific patterns and fails to generalize.

Interview WHY: How do you identify overfitting?

Dropout

Intuition: Randomly removes units during training to reduce co-adaptation.

Interview WHY: Why does dropout help generalization? Why is it disabled during inference?

L1 / L2 / Weight Decay

Intuition: Penalize parameter magnitude or control weight updates to reduce overfitting.

Interview WHY: What is the difference between L1 and L2 regularization?

Early Stopping

Intuition: Stops training when validation performance stops improving.

Interview WHY: Why monitor validation rather than training loss?

Data Augmentation

Intuition: Creates varied training examples that preserve the underlying label.

Interview WHY: Why does augmentation help computer vision models?

9. Batch Normalization and Normalization ■■■■

Batch Normalization

Intuition: Normalizes intermediate activations using batch statistics and learns scale and shift.

Interview WHY: Why can BatchNorm stabilize or speed up training? What changes during inference?

Layer Normalization

Intuition: Normalizes across features within an example and is common in Transformers.

Interview WHY: Why is LayerNorm preferred in many Transformer architectures?

10. Weight Initialization ■■■■

Xavier / Glorot

Intuition: Designed to keep activation/gradient variance reasonably controlled for certain activations.

Interview WHY: Why does initialization matter?

He Initialization

Intuition: Designed with ReLU-like activations in mind.

Interview WHY: Why is He initialization commonly paired with ReLU?

11. Vanishing and Exploding Gradients ■■■■

Vanishing Gradients

Intuition: Gradients become extremely small as they propagate through deep networks.

Interview WHY: Why do sigmoid/tanh and deep chains contribute to vanishing gradients?

Exploding Gradients

Intuition: Gradients become very large and make training unstable.

Interview WHY: How can gradient clipping help?

Solutions

Intuition: ReLU-family activations, good initialization, normalization, residual connections, and clipping can help.

Interview WHY: How do ResNets and initialization improve gradient flow?

12. CNNs ■■■■

Convolution

Intuition: A learned filter scans local regions and detects patterns.

Interview WHY: Why are convolutions effective for images?

Kernel / Filter

Intuition: Learned weights that detect features such as edges and textures.

Interview WHY: What does a convolution filter actually learn?

Feature Map

Intuition: Output produced by applying filters to an input.

Interview WHY: What does a feature map represent?

Stride and Padding

Intuition: Stride controls movement of the filter; padding controls border handling and output size.

Interview WHY: How do stride and padding affect feature-map dimensions?

Pooling

Intuition: Reduces spatial resolution and computation while retaining useful information.

Interview WHY: Why use max pooling? What information can pooling lose?

Receptive Field

Intuition: Region of the original input that influences a unit.

Interview WHY: Why does receptive field matter in object recognition?

Channels

Intuition: Separate feature dimensions such as RGB or learned feature maps.

Interview WHY: Why does channel count change across CNN layers?

13. CNN Architectures ■■■■

LeNet / AlexNet / VGG

Intuition: Know the historical progression and basic design ideas.

Interview WHY: What changed from early CNNs to deeper CNNs?

Inception

Intuition: Uses multiple receptive-field sizes in parallel.

Interview WHY: Why use multiple kernel sizes?

ResNet

Intuition: Uses skip connections so layers learn residual functions.

Interview WHY: Why do residual connections make very deep networks easier to train?

MobileNet

Intuition: Uses depthwise separable convolutions to reduce computation.

Interview WHY: Why is MobileNet useful for edge/mobile devices?

EfficientNet

Intuition: Balances depth, width, and resolution systematically.

Interview WHY: What is compound scaling?

14. Transfer Learning ■■■■

Feature Extraction

Intuition: Use a pretrained network as a fixed feature extractor.

Interview WHY: Why use a pretrained model instead of training from scratch?

Fine-tuning

Intuition: Unfreeze some or all pretrained layers and adapt them to the target dataset.

Interview WHY: When should you freeze layers and when should you fine-tune?

Pretrained Weights

Intuition: Weights learned from a large dataset provide reusable representations.

Interview WHY: Why can pretrained features generalize to a new dataset?

15. Object Detection ■■■■

Classification vs Detection

Intuition: Classification answers what; detection answers what and where.

Interview WHY: What is the difference between image classification and object detection?

Bounding Box

Intuition: Represents object location and size.

Interview WHY: How is a bounding box represented?

IoU

Intuition: Intersection over Union measures overlap between predicted and ground-truth boxes.

Interview WHY: Why is IoU important?

Confidence Score

Intuition: Model's estimated confidence that a predicted box contains an object/class.

Interview WHY: How is confidence used in detection?

Non-Maximum Suppression

Intuition: Removes redundant overlapping predictions and keeps stronger boxes.

Interview WHY: Why is NMS needed?

mAP

Intuition: Mean Average Precision summarizes detection precision-recall performance.

Interview WHY: What is mAP and why is it used for object detection?

YOLO

Intuition: Single-stage detection family designed for fast end-to-end detection.

Interview WHY: Why is YOLO fast? What is the difference between one-stage and two-stage detection?

16. Sequence Models ■■■■

RNN

Intuition: Maintains a hidden state to carry information across sequence steps.

Interview WHY: Why do ordinary feed-forward networks struggle with sequences?

LSTM

Intuition: Uses gates and a cell state to better preserve useful long-range information.

Interview WHY: Why was LSTM introduced? What do its gates do?

GRU

Intuition: Simplifies gated recurrent computation compared with LSTM.

Interview WHY: Why might GRU be chosen instead of LSTM?

Bidirectional RNN

Intuition: Processes sequence information from both directions.

Interview WHY: When is bidirectional context useful?

17. Attention ■■■■■■

Query, Key, Value

Intuition: Query asks what information is relevant; keys describe available information; values carry the information.

Interview WHY: What are Q, K, and V intuitively?

Self-Attention

Intuition: Each token can weigh other tokens to build a context-aware representation.

Interview WHY: Why is self-attention useful for long-range dependencies?

Scaled Dot-Product Attention

Intuition: Uses similarity between queries and keys to create attention weights over values.

Interview WHY: Why scale attention scores?

Multi-Head Attention

Intuition: Multiple attention heads learn different relationships in parallel.

Interview WHY: Why use multiple heads instead of one?

18. Transformers ■■■■■■

Transformer Architecture

Intuition: Uses attention and feed-forward blocks with residual connections and normalization.

Interview WHY: Why did Transformers become dominant for language tasks?

Positional Information

Intuition: Provides token-order information because attention alone does not inherently encode sequence order.

Interview WHY: Why do Transformers need positional information?

Encoder vs Decoder

Intuition: Encoder builds representations from input; decoder generates output autoregressively in decoder-only models.

Interview WHY: What is the difference between encoder-only, decoder-only, and encoder-decoder models?

Residual Connections

Intuition: Add the input of a block to its transformed output, helping information and gradients flow.

Interview WHY: Why are residual connections important in deep networks?

19. LLM Fundamentals ■■■■■■

Tokenization

Intuition: Converts text into tokens that the model processes.

Interview WHY: Why don't LLMs process raw sentences directly?

Embeddings

Intuition: Represent tokens or text as dense vectors capturing useful relationships.

Interview WHY: What is an embedding and why is it useful?

Pretraining

Intuition: Learns broad language/statistical patterns from large datasets.

Interview WHY: What does an LLM learn during pretraining?

Instruction Tuning / SFT

Intuition: Adapts a pretrained model to follow instructions using supervised examples.

Interview WHY: Why is instruction tuning needed?

RLHF

Intuition: Uses human preference signals to improve model behavior.

Interview WHY: Why use preference optimization after supervised fine-tuning?

Context Window

Intuition: Maximum amount of input/output context the model can consider in one request, depending on the model/API.

Interview WHY: What happens when relevant information exceeds the context window?

Temperature / Top-k / Top-p

Intuition: Control the randomness and diversity of token sampling.

Interview WHY: What happens when temperature is increased?

20. Fine-Tuning, LoRA and QLoRA ■■■■■■

Fine-tuning

Intuition: Adapts pretrained model parameters to a target task/domain.

Interview WHY: Why fine-tune instead of training from scratch?

LoRA

Intuition: Freezes the base model and trains small low-rank adapter matrices.

Interview WHY: Why does LoRA reduce trainable parameters and storage?

QLoRA

Intuition: Combines quantized base-model weights with LoRA adapters to reduce memory usage during fine-tuning.

Interview WHY: Why is QLoRA useful on limited-GPU hardware?

Quantization

Intuition: Represents model weights/activations with fewer bits to reduce memory and often improve inference efficiency.

Interview WHY: What is the trade-off between memory savings and model quality?

21. Embeddings and RAG ■■■■■■

Text Embeddings

Intuition: Map text into vectors so semantically similar text can be compared.

Interview WHY: Why use embeddings instead of exact keyword matching?

Cosine Similarity

Intuition: Measures angular similarity between vectors.

Interview WHY: Why is cosine similarity commonly used for text embeddings?

Vector Database

Intuition: Stores embeddings and supports similarity search.

Interview WHY: Why do we need a vector database in RAG?

Retrieval

Intuition: Finds relevant chunks for a user question.

Interview WHY: Why can retrieval quality determine RAG quality?

RAG

Intuition: Retrieval-Augmented Generation gives an LLM external retrieved context before generation.

Interview WHY: Why use RAG instead of putting all documents into a prompt?

Chunking

Intuition: Splits documents into manageable retrieval units.

Interview WHY: Why does chunk size affect RAG performance?

22. Generative Models ■■■■

Autoencoders

Intuition: Learn compressed representations and reconstruction.

Interview WHY: What is the purpose of the encoder and decoder?

VAE

Intuition: Learns a probabilistic latent representation.

Interview WHY: How does VAE differ from a standard autoencoder?

GAN

Intuition: Generator and discriminator compete in an adversarial setup.

Interview WHY: What are the roles of generator and discriminator?

Diffusion Models

Intuition: Learn to reverse a gradual noising process to generate data.

Interview WHY: Why does denoising enable generation?

23. Practical Deep Learning ■■■■■■

PyTorch Tensors

Intuition: Core numerical objects used to represent data and parameters.

Interview WHY: Why are tensors central to deep learning?

Dataset / DataLoader

Intuition: Separate data representation from efficient batching/shuffling/loading.

Interview WHY: Why use DataLoader instead of manually feeding data?

Training Loop

Intuition: Forward → loss → backward → optimizer step, repeated over batches.

Interview WHY: Explain a complete training iteration.

Validation Loop

Intuition: Measures generalization without updating parameters.

Interview WHY: Why don't we backpropagate through validation?

GPU / CUDA

Intuition: Accelerates tensor operations using GPU hardware.

Interview WHY: Why is GPU training faster for deep neural networks?

Checkpointing

Intuition: Saves model/optimizer state during training.

Interview WHY: Why save checkpoints?

Mixed Precision

Intuition: Uses lower precision where safe to improve speed and reduce memory.

Interview WHY: Why can mixed precision speed up training?

24. Deep Learning Evaluation ■■■■

Classification Metrics

Intuition: Accuracy, precision, recall, F1, ROC-AUC.

Interview WHY: Which metric would you choose for imbalanced data and why?

Object Detection Metrics

Intuition: IoU, precision, recall, mAP.

Interview WHY: Why is mAP more informative than accuracy for object detection?

Perplexity

Intuition: Measures how well a language model predicts a sequence; lower is generally better under comparable setups.

Interview WHY: What does perplexity intuitively measure?

BLEU / ROUGE

Intuition: Overlap-based metrics often used for generated text evaluation.

Interview WHY: Why can automatic text metrics fail to capture semantic quality?

■ Interview Priority Roadmap

Must Know: Neural network basics, forward propagation, loss, backpropagation, gradient descent, activation functions, learning rate, optimizers, overfitting/regularization.

Computer Vision: CNN, convolution, padding/stride, pooling, receptive field, transfer learning, object detection, IoU, NMS, mAP, YOLO, ResNet.

Modern AI / GenAI: Embeddings, attention, self-attention, Transformers, tokenization, context window, LLMs, fine-tuning, LoRA, QLoRA, quantization, RAG.

Practical: PyTorch tensors, Dataset/DataLoader, training/validation loops, GPU/CUDA, checkpoints, mixed precision.

■ Core Intuition Chain

Input → Weights + Bias → Activation → Prediction → Loss → Backpropagation → Gradients → Optimizer → Weight Update → Repeat

CNN → local patterns → edges → textures → shapes → objects

Attention → compare tokens → assign importance → combine relevant information

Transformer → self-attention + feed-forward + residual connections + normalization + positional information

RAG → document → chunks → embeddings → vector search → relevant context → LLM → grounded answer

Interview Rule

Do not only memorize definitions. For every important topic, be able to answer: What is it? Why do we need it? How does it work intuitively? What problem does it solve? What happens if we remove/change it? What are its limitations?